

Leader-following Consensus Control of Multi-Agent Systems with Communication Delays & Random Packet Loss

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Abstract—The objective of this paper is to develop a novel consensus control algorithm for multi-agent systems (MAS) experiencing constant and time-varying communication delays along with random packet data loss. The topology of the communication network of the system is modeled by an algebraic graph theory. The multi-agent dynamics is based on double integrator systems in discrete time domain. Bernoulli distribution principle is applied to system dynamics in order to deal with the uncertain packet data loss. The sufficient conditions for the stabilization of controller are formed by using Lyapunov-based methodologies and linear matrix inequalities (LMIs) techniques. The feasibility of the generated LMIs are analyzed to verify the stability of controller design. The control gain of the system is calculated if the solution of LMIs lie within the feasibility region, which ensures the MAS to achieve consensus. Simulation results with three agents are shown to verify the consensus of the proposed system.

I. INTRODUCTION

Multi-agent systems (MAS) are composed of several intelligent agents which are capable of interacting with each other through communication channels to solve some assigned complex tasks that are beyond the capabilities of a single agent. These agents may be any computer controlled entity that can be programmed and act accordingly across open and distributed environments. In our case, we are considering mobile robots as the agents. Recently, distributed coordinated control of multi-agent systems has attracted many researchers from control community to take part in its vast applications. Consensus control of multi-agent system is one of these applications which may be applied in areas of military demining, wireless collaboration and communications, military logistics planning, supply chain management, joint mission planning, oceanography, artificial intelligence, space-based combat, surveillance etc.

To enable these applications, various cooperative control capabilities need to be developed, including formation control, rendezvous, flocking, foraging, and air traffic control, etc. [1-2]. Consensus problems have recently been studied as in [3] with different control techniques. For example in [3], a distributed higher order consensus protocol is proposed, and the system can reach consensus if and only if all subsystems are asymptotically stable.

Since the MAS control is still considered as a relatively new area, a number of technical challenges need to be

overcome before they are to be used effectively. In [4], one of the most important challenges is that uncertainties caused by inter-vehicle communication, the information exchange among vehicles may be unreliable. Today, agents in multi-agent systems are mostly and popularly controlled through wireless networked systems. However, wireless networks are not always as reliable as hardwired ones due to connection strength, bandwidth constraints, which can cause packet delays and data loss. Therefore, the networked control system stability has been studied extensively associated with communication channel constraints, such as [5]. In [5], multi-agent systems are under continuous time-varying network and the dynamics of agents is described by a double-integrator type model. In [6], the dynamically changing interaction topologies are researched, which can represent some certain broken communication links due to bandwidth constraints. In [7], a random process is mentioned to model the communication networks with a time-varying topology, and showed that the connection weights in the communication network should be treated as control parameters. The challenge in consensus control is to find a control scheme which can allow the agreement of a group of agents on their states by interacting with each other. These states might be position, velocity, acceleration, temperature, torque and so on. Based on the literature works, the motivation of this work is to integrate the techniques in networked control systems for a single system to multi-agent system.

II. PROBLEM FORMULATION

A simple double integrator system in continuous time domain can be converted into discrete time domain by zero-order hold and sampler. The sampler is used to convert continuous signals into a train of amplitude modulated pulses and maintain the same value of pulse for a prescribed time duration. The converted discrete time double integrator system considering sampling time period as 0.1 second is

$$x_i(k+1) = A_d x_i(k) + B_d u_i(k) \quad (1)$$

where, $A_d = \begin{bmatrix} 0 & 0.1 \\ 0 & 0 \end{bmatrix}$, $B_d = \begin{bmatrix} 0.005 \\ 0.1 \end{bmatrix}$, k is the communication event, $A_d \in \mathbb{R}^{2 \times 2}$, $B_d \in \mathbb{R}^{2 \times 1}$ and $x_i \in \mathbb{R}$ and $u_i \in \mathbb{R}$ are the state and input matrices of agent i respectively.

The communication topology of a network of agents is represented using a directed graph also called as digraph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$. $\mathcal{V} = \{1, 2, \dots, N\}$ is the set of nodes and $\mathcal{E} = \{(i, j) : i, j \in \mathcal{V}\} \subset \mathcal{V} \times \mathcal{V}$ represent edges. Every digraph has associated with it an adjacency matrix $\mathcal{A}$ with non-negative binary elements. The order of adjacency matrix is

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$n \times n$, where n is the number of agents and is defined as $a_{ii} = 0$ and $a_{ij} \geq 0$. If there exists an edge between i and j , then the elements of matrix $\mathcal{A}$ is described as $a_{ij} > 0 \Leftrightarrow (i, j) \in \mathcal{E}$. The set of neighboring agents connected to any agent in topology is denoted by $\mathcal{N} = \{j \in \mathcal{V} : (i, j) \in \mathcal{E}\}$. The Laplacian matrix $\mathcal{L} = (l_{ij})$ of the digraph $\mathcal{G}$ is defined by $l_{ij} = -a_{ij}$ for $i \neq j$, $l_{ii} = -\sum_{j=1, j \neq i}^n l_{ij}$ ($i, j = 1, 2, \dots, N$).

Lemma 1 : The digraph $\mathcal{G}$ has a globally reachable node if and only if the Laplacian matrix $\mathcal{L}$ of $\mathcal{G}$ has a simple zero eigenvalue with associated eigenvector $1_n \triangleq [1, \dots, 1]^T \in \mathfrak{R}$.

The three agent communication topology under consideration is shown below in Fig.1. There is a leader agent among them and others are following agents or followers. The leader agent decides the trajectory to be followed in order to achieve consensus. It does so by receiving position coordinates of itself and followers and transmitting back the updated position co-ordinates to followers.



Fig. 1: Communication Topology of Three Agent System

In the communication topology shown in Fig. 1., agent A is the leader agent and agents B and C are followers. The adjacency matrix and the Laplacian matrix for this communication topology are,

$$\mathcal{A} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \text{ and } \mathcal{L} = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

The eigenvalues of the Laplacian matrix $\mathcal{L}$ are 0, 1 and 1. So, the simple zero eigenvalue indicates that the considered communication topology can reach consensus by graph theory. However, there might be some irregularities in system like packet data loss, time delays, switching topology and stochastic perturbations etc. which may deviate the multi-agent system from achieving consensus. Hence, it becomes necessary to design a state feedback controller which can guarantee consensus despite all irregularities present in the system.

Assumption 1 : The communication topology of the multi-agent system is static and non-switching i.e. at any given time within the communication network, the agents will receive and send information to same neighboring agents.

Assumption 2 : The communication time-delay between all communication edges is constant at a specific time. However, this may not be the scenario in real time condition. In order to tackle this problem, communication time-delay is classified into two cases, one of them being case of constant time-delay and other being time-varying delay case.

Case 1 : Sometimes, in networked control system, the time-delay may be infinitesimally small or there may not be much deviation from its average value. So it is better to consider the average value of time-delay as constant value of time-delay in controller design.

Case 2 : In some networked control systems, the difference between the maximum and minimum value of time-delay

may be of higher order than that of average value of time-delay. This situation affects the system behavior and may also affect the stability of system. So it becomes necessary to consider the upper and lower bounds of these time-varying delays in controller design to ensure system stability.

As discussed in earlier section, the basic objective of controller design is to reduce the error between the positions of the leader and the follower. The proposed discrete time controller (2) for the system (1) is given by

$$u_i(k) = K \sum_{j=1}^n a_{ij}(k)(x_j(k) - x_i(k)), \quad (2)$$

where $u_i(k)$ is the control input, K is the control gain of the system, $a_{ij}(k) \in \{0, 1\}$ is the (i, j) entry of an adjacency matrix $\mathcal{A} = [a_{ij}] \in \mathfrak{R}$ and $x_j(k)$ and $x_i(k)$ are the states of the neighboring agent and leader agent respectively. The closed loop multi-agent system (3) obtained from the state feedback controller is as follows,

$$x_i(k+1) = A_d x_i(k) + B_d K \sum_{j=1}^n a_{ij}(k)(x_j(k) - x_i(k)) \quad (3)$$

Definition 1 : The multi-agent system (3) is said to achieve leader-following consensus for any initial condition if it satisfies

$$\begin{cases} \lim_{k \rightarrow +\infty} \|x_j(k) - x_1(k)\| = 0, j = 2, 3, 4, \dots, N. \\ \lim_{k \rightarrow +\infty} \|\dot{x}_j(k) - \dot{x}_1(k)\| = 0, j = 2, 3, 4, \dots, N., \end{cases}$$

These conditions indicate that the difference or error between the position states of leader and follower should be zero as communication event or sample tends to infinity. However in real time condition, the minimum accepted error should be fixed to some appropriate value to achieve consensus in finite time. Deriving the error dynamics for the considered multi-agent system becomes necessary to satisfy these conditions.

In wireless networked control system, the agents transfer information from one to another in the form of data packets. These packets are transmitted and received in fixed time intervals also known as sampling time in discrete time system. Packet data loss is random because of numerous reasons like network function overload, weak router signal and so on. In modern wireless communication network, packet data loss is not a major concern. Still to ensure successful data transmission, it is necessary to consider packet data loss in controller design.

Lemma 2 : A Bernoulli process is considered to model the packet data loss in every time step. If there is a loss of packet at any time step, then the agent should trace data from previous time step to analyze its new position state. This process is random and memoryless but guarantees the latest successfully transmitted data. Bernoulli process combines data loss and delays into a single data loss case.

Assumption 3 : For an efficient data transmission, it is necessary to introduce a new factor which could establish a relationship between packet data losses and time delays. This factor is called as data loss rate ' r '. There must also

be a certain condition such that beyond the certain value of time-delay, if packet is received, then it must be considered as packet loss. This condition can only be determined by observing consensus time for specific data loss rate.

Definition 2 : For any transmitted data, the Bernoulli process is described as

$$x(k) = \theta(k)x(k) + (1 - \theta(k))x(k-1), \quad (4)$$

where $\theta(k) \in \mathfrak{R}$ is a stochastic variable satisfying Bernoulli distribution, i.e. $\theta \in \{0, 1\}$.

Assumption 4 : If the transmission between the agents is successful, then $\theta(k) = 1$ and if there is a packet loss, then $\theta(k) = 0$. In multi-agent system, the value of $\theta(k)$ depends on the follower agent. So it is always $\theta_{leader}(k) = \theta_{follower}(k)$. In (5), the value of $\theta_i(k)$ is same as $\theta_j(k)$. Hence it is better to replace $\theta_j(k)$ and $\theta_i(k)$ by $\theta(k)$.

After applying Bernoulli distribution (4) to the closed loop feedback multi-agent system (3), we get a new closed loop multi-agent system incorporating packet data loss.

$$\begin{aligned} x_i(k+1) &= A_d x_i(k) + B_d K \sum_{j=1}^n a_{ij}(k) [\theta(k)x_j(k) \\ &\quad + (1 - \theta(k))x_j(k-1) - \theta(k)x_i(k) \\ &\quad - (1 - \theta(k))x_i(k-1)] \end{aligned} \quad (5)$$

As discussed earlier in Case 1, the communication time-delays can be assumed constant in all communication edges at all times. The closed loop multi-agent system (5) experiencing a constant average time-delay of $\tau \in \mathfrak{R}$ can be expressed as (6),

$$\begin{aligned} x_i(k+1) &= A_d x_i(k) + B_d K \sum_{j=1}^n a_{ij}(k) [\theta(k)x_j(k-\tau) \\ &\quad + (1 - \theta(k))x_j(k-\tau-1) - \theta(k)x_i(k-\tau) \\ &\quad - (1 - \theta(k))x_i(k-\tau-1)] \end{aligned} \quad (6)$$

In the real time wireless network systems, time-delays can vary within a high range as discussed in Case 2. The upper and lower bounds of time-delays are always known in a wireless network. So it becomes necessary to consider an arbitrary variable value of time-delay τ_k which can take any values between the lower bound time delay τ_{min} and the upper bound time delay τ_{max} , such that $\tau_{min} \leq \tau_k \leq \tau_{max}$. The closed loop multi-agent system (5) experiencing a time-varying delay of $\tau_k \in \mathfrak{R}$ can be expressed as (7),

$$\begin{aligned} x_i(k+1) &= A_d x_i(k) + B_d K \sum_{j=1}^n a_{ij}(k) [\theta(k)x_j(k-\tau_k) \\ &\quad + (1 - \theta(k))x_j(k-\tau_k-1) - \theta(k)x_i(k-\tau_k) \\ &\quad - (1 - \theta(k))x_i(k-\tau_k-1)] \end{aligned} \quad (7)$$

Equations (6) and (7) represent multi-agent system with constant time-delay and time-varying delay respectively for a single agent i . The general equations (8) and (9) equivalent to (6) and (7) respectively for n agents and can be expressed in matrix form which is as follows,

$$\begin{aligned} x(k+1) &= (I_{n \times n} \otimes A_d)x(k) + (L_1 \otimes B_d K)x(k-\tau) \\ &\quad + (L_2 \otimes B_d K)x(k-\tau-1) \end{aligned} \quad (8)$$

$$\begin{aligned} x(k+1) &= (I_{n \times n} \otimes A_d)x(k) + (L_1 \otimes B_d K)x(k-\tau_k) \\ &\quad + (L_2 \otimes B_d K)x(k-\tau_k-1) \end{aligned} \quad (9)$$

where $\otimes$ is the kronecker product, $x = [x_1, x_2, \dots, x_n]^T$, $L_1 \in \mathfrak{R}$ and $L_2 \in \mathfrak{R}$. The values of A_d , L_1 and L_2 are independent of both constant and time-varying delay conditions. These matrices are dependent on adjacency matrix elements and packet data loss variable.

$$L_1 = \begin{bmatrix} -\sum_{j=1}^n a_{1j}\theta & a_{12}\theta & \dots & a_{1n}\theta \\ a_{21}\theta & -\sum_{j=1}^n a_{2j}\theta & \dots & a_{2n}\theta \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}\theta & a_{n2}\theta & \dots & -\sum_{j=1}^n a_{nj}\theta \end{bmatrix} \quad (10)$$

$$L_2 = \begin{bmatrix} -\sum_{j=1}^n a_{1j}(1-\theta) & a_{12}(1-\theta) & \dots & a_{1n}(1-\theta) \\ a_{21}(1-\theta) & -\sum_{j=1}^n a_{2j}(1-\theta) & \dots & a_{2n}(1-\theta) \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}(1-\theta) & a_{n2}(1-\theta) & \dots & -\sum_{j=1}^n a_{nj}(1-\theta) \end{bmatrix} \quad (11)$$

The main objective of achieving consensus is to design a state feedback controller or control gain to reduce the error. Also, for designing a controller, error dynamics is required as an energy function in assuming a Lyapunov function. The error dynamics for multi-agent system with constant time delay (8) and for time-varying delay (9) is given by (12) and (13) respectively. Here $e_i(k) = x_i(k) - x_1(k)$.

$$\begin{aligned} e_i(k+1) &= A_d e_i(k) + B_d K \sum_{j=2}^n (a_{ij} - a_{1j})\theta(k)e_j(k-\tau) \\ &\quad + B_d K \sum_{j=2}^n (a_{ij} - a_{1j})(1-\theta(k))e_j(k-\tau-1) \\ &\quad - B_d K \sum_{j=1}^n (a_{ij} - a_{1j})\theta(k)e_i(k-\tau) \\ &\quad - B_d K \sum_{j=1}^n (a_{ij} - a_{1j})(1-\theta(k))e_i(k-\tau-1) \end{aligned} \quad (12)$$

$$\begin{aligned} e_i(k+1) &= A_d e_i(k) + B_d K \sum_{j=2}^n (a_{ij} - a_{1j})\theta(k)e_j(k-\tau_k) \\ &\quad + B_d K \sum_{j=2}^n (a_{ij} - a_{1j})(1-\theta(k))e_j(k-\tau_k-1) \\ &\quad - B_d K \sum_{j=1}^n (a_{ij} - a_{1j})\theta(k)e_i(k-\tau_k) \\ &\quad - B_d K \sum_{j=1}^n (a_{ij} - a_{1j})(1-\theta(k))e_i(k-\tau_k-1) \end{aligned} \quad (13)$$

For n number of agents, the general multi-agent error dynamics for constant time-delay error dynamics (12) and time-varying delay error dynamics (13) is shown in (14) and (15). Here $e(k) = [e_2(k), \dots, e_n(k)]^T$.

$$e(k+1) = (I_{n-1 \times n-1} \otimes A_d)e(k) + (\bar{L}_1 \otimes B_d K)e(k - \tau) + (\bar{L}_2 \otimes B_d K)e(k - \tau - 1) \quad (14)$$

$$e(k+1) = (I_{n-1 \times n-1} \otimes A_d)e(k) + (\bar{L}_1 \otimes B_d K)e(k - \tau_k) + (\bar{L}_2 \otimes B_d K)e(k - \tau_k - 1) \quad (15)$$

where $\bar{L}_1 \in \Re$ and $\bar{L}_2 \in \Re$. The values of A_d , $\bar{L}_1$ and $\bar{L}_2$ are independent of both constant and time-varying delay conditions. These matrices are dependent on adjacency matrix elements and packet data loss variable.

$$\bar{L}_1 = \begin{bmatrix} (a_{22} - a_{12})\theta - O_1 & (a_{23} - a_{13})\theta & \dots & (a_{2n} - a_{1n})\theta \\ (a_{32} - a_{12})\theta & (a_{33} - a_{13})\theta - O_2 & \dots & (a_{3n} - a_{1n})\theta \\ \vdots & \vdots & \ddots & \vdots \\ (a_{n2} - a_{12})\theta & (a_{n3} - a_{13})\theta & \dots & (a_{nn} - a_{1n})\theta - O_3 \end{bmatrix} \quad (16)$$

$$\bar{L}_2 = \begin{bmatrix} (a_{22} - a_{12})S_1 - Q_1 & (a_{23} - a_{13})S_2 & \dots & (a_{2n} - a_{1n})S_n \\ (a_{32} - a_{12})S_1 & (a_{33} - a_{13})S_2 - Q_2 & \dots & (a_{3n} - a_{1n})S_n \\ \vdots & \vdots & \ddots & \vdots \\ (a_{n2} - a_{12})S_1 & (a_{n3} - a_{13})S_2 & \dots & (a_{nn} - a_{1n})S_n - O_3 \end{bmatrix} \quad (17)$$

where $n \geq 2$, $O_1 = \sum_{j=1}^n a_{2j}\theta$, $O_2 = \sum_{j=1}^n a_{3j}\theta$, $O_3 = \sum_{j=1}^n a_{nj}\theta$, $Q_1 = \sum_{j=1}^n a_{2j}S_j$, $Q_2 = \sum_{j=1}^n a_{3j}S_j$, $Q_3 = \sum_{j=1}^n a_{nj}S_j$ and $S = (1 - \theta)$.

III. MAIN RESULT

In this section, the stability analysis of the controller design is verified by using linear matrix inequalities (LMIs). The control gain design for the multi-agent control system is carried out by using Lyapunov-based analysis. LMIs are applied in estimation, identification, matrix scaling problems and so on. The main strength of LMIs is the ability to combine various designs constraints or objectives in a numerically tractable manner.

Definition 3 : A Lyapunov function $V : \Re^n \rightarrow \Re$ is positive definite if $V(z) \geq 0$ for all z or $V(z) = 0$ if and only if $z = 0$ or all sublevel sets of V are bounded.

Theorem 1 : For the multi-agent system (8), if there exists positive definite matrices $\hat{P}$, $\hat{Q}$, $\hat{R}$ and a symmetric matrix $M_1 = M_1^T$ and also if the following linear matrix inequality (18) holds true, then the control gain K is designed as $K = X^{-1}Y$ with $X = M_1^{-1} \in \Re$ and $Y \in \Re$.

$$\begin{bmatrix} \Phi_{11} & \Phi_{21}^T & \Phi_{31}^T & \Phi_{41}^T & (\tau+1)\hat{N}_1 \\ \Phi_{21} & \Phi_{22} & \Phi_{32}^T & \Phi_{42}^T & (\tau+1)\hat{N}_2 \\ \Phi_{31} & \Phi_{32} & \Phi_{33} & \Phi_{43}^T & (\tau+1)\hat{N}_3 \\ \Phi_{41} & \Phi_{42} & \Phi_{43} & \Phi_{44} & (\tau+1)\hat{N}_4 \\ (\tau+1)\hat{N}_1^T & (\tau+1)\hat{N}_2^T & (\tau+1)\hat{N}_3^T & (\tau+1)\hat{N}_4^T & -(\tau+1)\hat{R} \end{bmatrix} < 0 \quad (18)$$

where,

$$\begin{aligned} \Phi_{11} &= r\hat{P} + (\tau+1)(1-r)\hat{R} + \hat{N}_1 + \hat{N}_1^T + 2X \\ \Phi_{21} &= -(\tau+1)^T \hat{R}^T + \hat{N}_2 - T_1^T X + \omega_1 X \end{aligned}$$

$$\begin{aligned} \Phi_{22} &= -(1-r)\hat{P} + r\hat{Q} + (\tau+1)r\hat{R} - \omega_1 T_1 X - \omega_1 T_1^T X \\ \Phi_{31} &= \hat{N}_3 - \hat{N}_1^T - T_2^T Y \\ \Phi_{32} &= -\hat{N}_2^T - \omega_1 T_2^T Y - \omega_2 T_1 X \\ \Phi_{33} &= -(1-r)\hat{Q} - \hat{N}_3 - \hat{N}_3^T - \omega_2 T_2 Y - \omega_2 T_2^T Y \\ \Phi_{41} &= -T_3 Y + \omega_3 X \\ \Phi_{42} &= -\omega_1 T_3^T Y - \omega_3 T_3 Y \\ \Phi_{43} &= -\omega_2 T_3^T Y - \omega_3 T_2 Y \\ \Phi_{44} &= -\omega_3 T_3 Y - \omega_3 T_3^T Y \\ \hat{P} &= M_1^{-1} P M_1^{-1} \\ \hat{Q} &= M_1^{-1} Q M_1^{-1} \\ \hat{R} &= M_1^{-1} R M_1^{-1} \\ \hat{S} &= M_1^{-1} S M_1^{-1} \\ \hat{N}_1 &= M_1^{-1} N_1 M_1^{-1} \\ \hat{N}_2 &= M_1^{-1} N_2 M_1^{-1} \\ \hat{N}_3 &= M_1^{-1} N_3 M_1^{-1} \\ \hat{N}_4 &= M_1^{-1} N_4 M_1^{-1} \end{aligned}$$

Proof : A Lyapunov function is chosen as,

$$V(k) = V_1(k) + V_2(k) + V_3(k) \quad (19)$$

where,

$$V_1(k) = e^T(k) P e(k)$$

$$V_2(k) = \sum_{i=1}^{\tau} e^T(k-i) Q e(k-i)$$

$$V_3(k) = \sum_{i=-\tau}^0 \sum_{j=i+k}^{k-1} (e^T(j+1) - e^T(j)) R (e(j+1) - e(j))$$

If it can be shown that $\Delta V = V(e, k+1) - V(e, k) < 0$, then $e(k)$ converges asymptotically to zero. In terms of data loss rate, matrices P , Q , and R can be written as (rP) , (rQ) and (rR) for their current sample and $((1-r)P)$, $((1-r)Q)$ and $((1-r)R)$ for the previous sample respectively. After substituting error dynamics in ΔV , we get individual values of ΔV_1 , ΔV_2 and ΔV_3 .

Apart from ΔV_1 , ΔV_2 and ΔV_3 , there are two zero-equations ψ_0 and ψ_1 introduced to eliminate undesired summation terms.

$$\psi_0 = (\tau+1)z^T(k)NR^{-1}N^T z(k) = 0 \quad (20)$$

$$\psi_1 = z^T(k) \begin{bmatrix} \pi_1 & \pi_2 \\ \pi_3 & \pi_4 \end{bmatrix} z(k) = 0 \quad (21)$$

Then $\Delta V = \Delta V_1 + \Delta V_2 + \Delta V_3 + \psi_0 + \psi_1$

$$\Delta V = z^T(k) \begin{bmatrix} \Xi_{11} & \Xi_{21}^T & \Xi_{31}^T & \Xi_{41}^T \\ \Xi_{21} & \Xi_{22} & \Xi_{32}^T & \Xi_{42}^T \\ \Xi_{31} & \Xi_{32} & \Xi_{33} & \Xi_{43}^T \\ \Xi_{41} & \Xi_{42} & \Xi_{43} & \Xi_{44} \end{bmatrix} z(k) + (\tau+1)z^T(k)NR^{-1}N^T z(k) \quad (22)$$

where,

$$\begin{aligned} \Xi_{11} &= rP + (\tau+1)rR + N_1 + N_1^T + M_1 + M_1^T \\ \Xi_{21} &= -(\tau+1)(1-r)R^T + N_2 - M_1^T T_1^T + \omega_1 M_1 \\ \Xi_{22} &= -(1-r)P + rQ + (\tau+1)rR - \omega_1 M_1 T_1 - T_1^T \omega_1 M_1^T \\ \Xi_{31} &= N_3 - N_1^T - M_1^T T_2^T K^T \\ \Xi_{32} &= -N_2^T - \omega_1 M_1^T T_2^T K^T - T_1 \omega_2 M_1 \\ \Xi_{33} &= -(1-r)Q - N_3 - N_3^T - \omega_2 M_1 T_2 K - T_2^T K^T \omega_2 M_1^T \\ \Xi_{41} &= -M_1^T T_3^T K^T + \omega_3 M_1 \\ \Xi_{42} &= -\omega_1 M_1^T T_3^T K^T - T_1 \omega_3 M_1 \end{aligned}$$

$$\begin{aligned}\Xi_{43} &= -\omega_2 M_1^T T_3^T K^T - T_2 K \omega_3 M_1 \\ \Xi_{44} &= -\omega_3 M_1 T_3 K - T_3^T K^T \omega_3 M_1^T\end{aligned}$$

Definition 4 : The Schur complement for any matrix X of the form $X = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, gives the following condition,

$$X < 0 \Leftrightarrow A - BD^{-1}C < 0$$

The Schur complement gives new equation, such that if it holds true, then the error will be bounded, i.e. $\Delta V < 0$, which indirectly means that the matrix $\Xi < 0$, which is expressed as in below equation.

$$\Xi = \begin{bmatrix} \Xi_{11} & \Xi_{21} & \Xi_{31} & \Xi_{41} & (\tau+1)N_1 \\ \Xi_{21} & \Xi_{22} + S & \Xi_{32} & \Xi_{42} & (\tau+1)N_2 \\ \Xi_{31} & \Xi_{32} & \Xi_{33} & \Xi_{43} & (\tau+1)N_3 \\ \Xi_{41} & \Xi_{42} & \Xi_{43} & \Xi_{44} & (\tau+1)N_4 \\ (\tau+1)N_1^T & (\tau+1)N_2^T & (\tau+1)N_3^T & (\tau+1)N_4^T & -(\tau+1)R \end{bmatrix} < 0 \quad (23)$$

LMI (23) contains arbitrary symmetric positive definite matrices P, Q, R, M and N . If the values for these matrices can be found out by calculated control gain, then we can say that the considered system will be stable. It is necessary to linearize the above inequality for which we need to pre and post multiply by M_1^{-1} . Since M_1 is a symmetric matrix, $M_1^T = M_1$. Assume $X = M_1^{-1}$, which is also a symmetric matrix and $Y = KX$, we get,

$$\Phi = M_1^{-1} \cdot \Xi \cdot M_1^{-1} = \begin{bmatrix} \Phi_{11} & \Phi_{21}^T & \Phi_{31}^T & \Phi_{41}^T & (\tau+1)\hat{N}_1 \\ \Phi_{21} & \Phi_{22} & \Phi_{32}^T & \Phi_{42}^T & (\tau+1)\hat{N}_2 \\ \Phi_{31} & \Phi_{32} & \Phi_{33} & \Phi_{43}^T & (\tau+1)\hat{N}_3 \\ \Phi_{41} & \Phi_{42} & \Phi_{43} & \Phi_{44} & (\tau+1)\hat{N}_4 \\ (\tau+1)\hat{N}_1^T & (\tau+1)\hat{N}_2^T & (\tau+1)\hat{N}_3^T & (\tau+1)\hat{N}_4^T & -(\tau+1)\hat{R} \end{bmatrix} < 0 \quad (24)$$

Theorem 2 : For the multi-agent system (9), if there exists positive definite matrices $\hat{P}, \hat{Q}, \hat{R}$ and a symmetric matrix $M_1 = M_1^T$ and also if the following linear matrix inequality (18) holds true, then the control gain K is designed as $K = X^{-1}Y$ with $X = M_1^{-1} \in \Re$ and $Y \in \Re$.

$$\Phi = M_1^{-1} \cdot \Xi \cdot M_1^{-1} = \begin{bmatrix} \Phi_{11} & \Phi_{21}^T & \Phi_{31}^T & \Phi_{41}^T & (\tau_M+1)\hat{N}_1 \\ \Phi_{21} & \Phi_{22} & \Phi_{32}^T & \Phi_{42}^T & (\tau_M+1)\hat{N}_2 \\ \Phi_{31} & \Phi_{32} & \Phi_{33} & \Phi_{43}^T & (\tau_M+1)\hat{N}_3 \\ \Phi_{41} & \Phi_{42} & \Phi_{43} & \Phi_{44} & (\tau_M+1)\hat{N}_4 \\ (\tau_M+1)\hat{N}_1^T & (\tau_M+1)\hat{N}_2^T & (\tau_M+1)\hat{N}_3^T & (\tau_M+1)\hat{N}_4^T & -(\tau_M+1)\hat{R} \end{bmatrix} < 0 \quad (25)$$

where,

$$\begin{aligned}\Phi_{11} &= r\hat{P} + (\tau_M+1)(1-r)\hat{R} + \hat{N}_1 + \hat{N}_1^T + 2X \\ \Phi_{21} &= -(\tau_M+1) \cdot (1-r)\hat{R}^T + \hat{N}_2 - T_1^T X + \omega_1 X \\ \Phi_{22} &= -(1-r)\hat{P} + (\tau_M - \tau_m + 1)\hat{Q} + (\tau_M+1)r\hat{R} - \omega_1 T_1 X - \omega_1 T_1^T X \\ \Phi_{31} &= \hat{N}_3 - \hat{N}_1^T - T_2^T Y \\ \Phi_{32} &= -\hat{N}_2^T - \omega_1 T_2^T Y - \omega_2 T_1 X \\ \Phi_{33} &= -(1-r)\hat{Q} - \hat{N}_3 - \hat{N}_3^T - \omega_2 T_2 Y - \omega_2 T_2^T Y \\ \Phi_{41} &= -T_3 Y + \omega_3 X \\ \Phi_{42} &= -\omega_1 T_3^T Y - \omega_3 T_3 Y \\ \Phi_{43} &= -\omega_2 T_3^T Y - \omega_3 T_2 Y \\ \Phi_{44} &= -\omega_3 T_3 Y - \omega_3 T_3^T Y \\ \tau_M &= \text{Upper bound time-delay.} \\ \tau_m &= \text{Lower bound time-delay.}\end{aligned}$$

Proof : Let us assume the following Lyapunov function for the system comprising of time-varying delays and packet data loss as,

$$V(k) = V_1(k) + V_2(k) + V_3(k) + V_4(k)$$

where,

$$V_1(k) = e^T(k)Pe(k)$$

$$V_2(k) = \sum_{i=k-\tau_k}^{k-1} e^T(i)Qe(i)$$

$$V_3(k) = \sum_{j=-\tau_M+1}^{-\tau_m+1} \sum_{i=k+j-1}^{k-1} e^T(i)Qe(i)$$

$$V_4(k) = \sum_{i=-\tau_M}^{-1} \sum_{j=k+1}^{k-1} (e^T(j+1) - e^T(j))R(e(j+1) - e(j))$$

Similarly to Theorem 1., the individual values of $\Delta V_1, \Delta V_2, \Delta V_3$ and ΔV_4 are calculated.

$\Delta V = \Delta V_1 + \Delta V_2 + \Delta V_3 + \Delta V_4 + \psi_0 + \psi_1$, where ψ_0, ψ_1 have the same values as that in constant time delay case. So adding up all matrices, we get ΔV as

$$\Delta V = z^T(k) \begin{bmatrix} \Xi_{11} & \Xi_{21}^T & \Xi_{31}^T & \Xi_{41}^T \\ \Xi_{21} & \Xi_{22} & \Xi_{32}^T & \Xi_{42}^T \\ \Xi_{31} & \Xi_{32} & \Xi_{33} & \Xi_{43}^T \\ \Xi_{41} & \Xi_{42} & \Xi_{43} & \Xi_{44} \end{bmatrix} z(k) + (\tau_M+1)z^T(k)NR^{-1}N^T z(k) \quad (26)$$

where,

$$\begin{aligned}\Xi_{11} &= rP + (\tau_M+1)rR + N_1 + N_1^T + M_1 + M_1^T \\ \Xi_{21} &= -(\tau_M+1)(1-r)R^T + N_2 - M_1^T T_1^T + \omega_1 M_1 \\ \Xi_{22} &= -(1-r)P + (\tau_M - \tau_m + 1)Q + (\tau_M+1)rR - \omega_1 M_1 T_1 - T_1^T \omega_1 M_1^T \\ \Xi_{31} &= N_3 - N_1^T - M_1^T T_2^T K^T \\ \Xi_{32} &= -N_2^T - \omega_1 M_1^T T_2^T K^T - T_1 \omega_2 M_1 \\ \Xi_{33} &= -Q - N_3 - N_3^T - \omega_2 M_1 T_2 K - T_2^T K^T \omega_2 M_1^T \\ \Xi_{41} &= -M_1^T T_3^T K^T + \omega_3 M_1 \\ \Xi_{42} &= -\omega_1 M_1^T T_3^T K^T - T_1 \omega_3 M_1 \\ \Xi_{43} &= -\omega_2 M_1^T T_3^T K^T - T_2 K \omega_3 M_1 \\ \Xi_{44} &= -\omega_3 M_1 T_3 K - T_3^T K^T \omega_3 M_1^T\end{aligned}$$

Applying Schur Complement on (26), we get a new equation in form,

$$\Xi = \begin{bmatrix} \Xi_{11} & \Xi_{21}^T & \Xi_{31}^T & \Xi_{41}^T & (\tau_M+1)N_1 \\ \Xi_{21} & \Xi_{22} + S & \Xi_{32}^T & \Xi_{42}^T & (\tau_M+1)N_2 \\ \Xi_{31} & \Xi_{32} & \Xi_{33} & \Xi_{43}^T & (\tau_M+1)N_3 \\ \Xi_{41} & \Xi_{42} & \Xi_{43} & \Xi_{44} & (\tau_M+1)N_4 \\ (\tau_M+1)N_1^T & (\tau_M+1)N_2^T & (\tau_M+1)N_3^T & (\tau_M+1)N_4^T & -(\tau_M+1)R \end{bmatrix} < 0 \quad (27)$$

The variables in this Ξ matrix are same as that derived Ξ matrix in constant-time delay case. If the values for these matrices can be found out by calculated control gain, then we can say that the considered system will be stable. It is necessary to linearize the above inequality for which we need to pre and post multiply by M_1^{-1} . Since M_1 is a symmetric matrix, $M_1^T = M_1$. Assume $X = M_1^{-1}$, which is also a symmetric matrix and $Y = KX$, we get,

$$\Phi = M_1^{-1} \cdot \Xi \cdot M_1^{-1} = \begin{bmatrix} \Phi_{11} & \Phi_{21}^T & \Phi_{31}^T & \Phi_{41}^T & (\tau_M+1)\hat{N}_1 \\ \Phi_{21} & \Phi_{22} & \Phi_{32}^T & \Phi_{42}^T & (\tau_M+1)\hat{N}_2 \\ \Phi_{31} & \Phi_{32} & \Phi_{33} & \Phi_{43}^T & (\tau_M+1)\hat{N}_3 \\ \Phi_{41} & \Phi_{42} & \Phi_{43} & \Phi_{44} & (\tau_M+1)\hat{N}_4 \\ (\tau_M+1)\hat{N}_1^T & (\tau_M+1)\hat{N}_2^T & (\tau_M+1)\hat{N}_3^T & (\tau_M+1)\hat{N}_4^T & -(\tau_M+1)\hat{R} \end{bmatrix} < 0 \quad (28)$$

Note that the proof in Theorem 1 and 2 are simplified due to limited space.

IV. NUMERICAL SIMULATIONS

In this section, numerical simulation results are shown to discuss the consensus control of three agents. The network topology under consideration is same as that shown in Fig 1.

For constant delay case, the control gain K is obtained for different data loss rate ranging from 0% to 98%. It was observed that the agents achieve consensus most quickly at data loss rate of 30%. The initial value of agents 1, 2 and 3 were 5, 10 and 15 meters respectively i.e. all agents are kept at a distance of 5 meters from each other. From Fig 2., it can be observed that the consensus time was observed around 19 seconds.

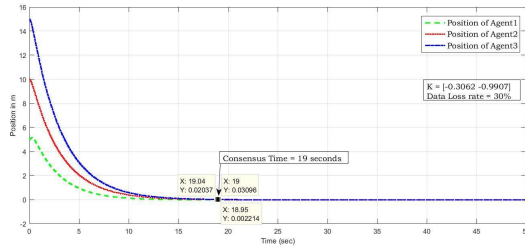


Fig. 2: Position of three agents for 30% Data Loss Rate

It was observed that as data loss rate increases till 30%, the consensus time decreases. Increasing data loss rate after 30% results in increase in consensus time. For practical purpose, the finite time for consensus should not exceed 50 seconds. Multi-agent system with data loss rate of 40% were able to reach consensus within 50 seconds. However increase in data loss rate exceeded this finite time boundary. This is equivalent to not achieving consensus.

Communication weight is another factor which can be introduced in networked control systems to control the data flow rate between the agents. Communication weight in multi-agent system helps in achieving consensus between any pair of agents faster or slower than other pair. In (18), ω_1 , ω_2 and $\omega_3 \in (0,1)$ are the communication weights. The values of ω_1 , ω_2 and ω_3 were changed and different cases were studied.

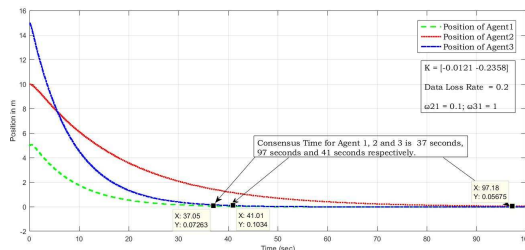


Fig. 3: Position for 20% Data Loss Rate and $\omega_{31} = 1$, $\omega_{21} = 0.1$

Substituting the values of $\tau_m = 0.01$ second and $\tau_M = 1$ in LMI (28) yields the value of control gain $K = [-0.3312 -0.9679]$. The calculated control gain confirms consensus of

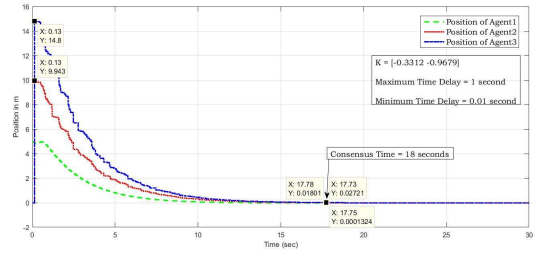


Fig. 4: Position for 30% Data Loss Rate and $\tau_m = 0.01$ second and $\tau_M = 1$ second

all agents in around 18 seconds as shown in Fig 4., which is faster than consensus of multi-agent system with same data loss rate and constant time-delay.

V. CONCLUSIONS

The leader-following consensus problem for double-integrator multi-agent systems with constant time-delays and time-varying delays and random packet data loss is investigated in this paper. Bernoulli distribution process is used to model the packet data loss in the multi-agent system. A multi-agent system with disturbances like packet data loss and time-delays can still achieve leader-following consensus by using the common Lyapunov function and linear matrix inequality method. From simulation results, its been observed that there is not a linear relationship between consensus time and disturbances present in the system. However, the desired objective of the research work will always be achieving consensus in minimum possible time at minimal cost of disturbances.

REFERENCES

- [1] R. M. Murray, "Recent Research in Cooperative Control of Multivehicle systems," Journal of Dynamic System, Measurement, and Control, vol. 129, 571-583, 2007.
- [2] J. Lu , G. Chen and X. Yu, "Modeling, analysis and control of multi-agent systems:A brief overview," in Proc. 2011 IEEE Int. Symp.Circuits Syst., pp. 2103-2106, 2011.
- [3] W. Yu , G. Chen , W. Ren , J. Kurths and W. Zheng "Distributed higher order consensus protocols in multi-agent dynamical systems", IEEE Trans. Circuits Syst. I, Reg. Papers, vol. 58, no. 8, pp. 1924-1932, 2011.
- [4] Wei Ren, Beard R.W., "Consensus seeking in multiagent systems under dynamically changing interaction topologies," Automatic Control, IEEE Transactions on , vol.50, no.5, pp. 655- 661, 2005.
- [5] Y. Zheng, H. Shao and W. Pan, "Consensus problem of double-integrator dynamics system under time-varying networks," Asian Control Conference, pp.343-348, 2009.
- [6] M.Yu, L. Wang, T.Chu and F.Hao, "An LMI approach to networked control systems with data packet dropout and transmission delays," Decision and Control, 2004. CDC. 43rd IEEE Conference on , vol.4, pp. 3545- 3550 Vol.4, 2004.
- [7] Y. Zhang and Y. P. Tian, "Consensus in Networked Multi-Agent Systems With Random Packet Loss," International Conference on Physics and Control, Catania, Italy, 2009.